J. Dairy Sci. 106:4559–4579
https://doi.org/10.3168/jds.2022-22334
© 2023, The Authors. Published by Elsevier Inc. and Fass Inc. on behalf of the American Dairy Science Association®.
This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Rumen-protected choline and methionine during the periparturient period affect choline metabolites, amino acids, and hepatic expression of genes associated with one-carbon and lipid metabolism

S. B. Potts,[1]* K. M. Brady,[2] C. M. Scholte,[3] K. M. Moyes,[3] N. E. Sunny,[3] and R. A. Erdman[3]
[1]Western Maryland Research and Education Center, University of Maryland Extension, Keedysville 21756
[2]Animal Biosciences and Biotechnology Laboratory, Beltsville Agricultural Research Center, Agricultural Research Service, USDA, Beltsville, MD 20705
[3]Department of Animal and Avian Sciences, University of Maryland, College Park 20742

ABSTRACT

Feeding supplemental choline and Met during the periparturient period can have positive effects on cow performance; however, the mechanisms by which these nutrients affect performance and metabolism are unclear. The objective of this experiment was to determine if providing rumen-protected choline, rumen-protected Met, or both during the periparturient period modifies the choline metabolitic profile of plasma and milk, plasma AA, and hepatic mRNA expression of genes associated with choline, Met, and lipid metabolism. Cows (25 primiparous, 29 multiparous) were blocked by expected calving date and parity and randomly assigned to 1 of 4 treatments: control (no rumen-protected choline or rumen-protected Met); CHO (13 g/d choline ion); MET (9 g/d DL-methionine prepartum; 13.5 g/d DL-methionine, postpartum); or CHO + MET. Treatments were applied daily as a top dress from ~21 d prepartum through 35 d in milk (DIM). On the day of treatment enrollment (d −19 ± 2 relative to calving), blood samples were collected for covariate measurements. At 7 and 14 DIM, samples of blood and milk were collected for analysis of choline metabolites, including 16 species of phosphatidylcholine (PC) and 4 species of lysophosphatidylcholine (LPC). Blood was also analyzed for AA concentrations. Liver samples collected from multiparous cows on the day of treatment enrollment and at 7 DIM were used for gene expression analysis. There was no consistent effect of CHO or MET on milk or plasma free choline, betaine, sphingomyelin, or glycerophosphocholine. However, CHO increased milk secretion of total LPC irrespective of MET for multiparous cows and in absence of MET for primiparous cows. Furthermore, CHO increased or tended to increase milk secretion of LPC 16:0, LPC 18:1, and LPC 18:0 for primi- and multiparous cows, although the response varied with MET supplementation. Feeding CHO also increased plasma concentrations of LPC 16:0 and LPC 18:1 in absence of MET for multiparous cows. Although milk secretion of total PC was unaffected, CHO and MET increased secretion of 6 and 5 individual PC species for multiparous cows, respectively. Plasma concentrations of total PC and individual PC species were unaffected by CHO or MET for multiparous cows, but MET reduced total PC and 11 PC species during wk 2 postpartum for primiparous cows. Feeding MET consistently increased plasma Met concentrations for both primi- and multiparous cows. Additionally, MET decreased plasma serine concentrations during wk 2 postpartum and increased plasma phenylalanine in absence of CHO for multiparous cows. In absence of MET, CHO tended to increase hepatic mRNA levels of betaine-homocysteine methyltransferase and phosphate cytidylyltransferase 1 choline, α, but tended to decrease expression of 3-hydroxy-3-methylglutaryl-coenzyme A synthase 2 and peroxisome proliferator activated receptor α irrespective of MET. Although shifts in the milk and plasma PC profile were subtle and inconsistent between primi- and multiparous cows, gene expression results suggest that supplemental choline plays a probable role in promoting the cytidine diphosphate-choline and betaine-homocysteine S-methyltransferase pathways. However, interactive effects suggest that this response depends on Met availability, which may explain the inconsistent results observed among studies when supplemental choline is fed.

Key words: transition cow, choline, methionine

INTRODUCTION

The choline requirement for dairy cattle has not been defined (NRC, 2001), although it is considered to be an essential nutrient for other mammals, including humans (Zeisel and da Costa, 2009). Choline deficiency impairs an animal's ability to export lipids from the

Received May 24, 2022.
Accepted December 29, 2022.
*Corresponding author: sbpotts@umd.edu

liver because it is required for synthesis of phosphatidylcholine (**PC**), the major phospholipid that comprises very-low-density lipoproteins (Yao and Vance, 1988). Phosphatidylcholine is synthesized through one of 2 pathways: (1) the cytidine diphosphate (**CDP**)-choline pathway using choline derived from the diet; and (2) the phosphatidylethanolamine N-methyltransferase (PEMT) pathway whereby a series of 3 methylation reactions occur to convert phosphatidylethanolamine to PC (Caudill, 2010). Because dietary choline is rapidly degraded in the rumen (Atkins et al., 1988), it is likely that in ruminants the majority of PC is synthesized via the PEMT pathway. Feeding rumen-protected choline (**RPC**) could lead to more PC synthesis via the CDP-choline pathway. DeLong et al. (1999) suggested that PC species that contained PUFA were derived from PEMT origin in rats. Thus, the profile of individual PC species in blood or milk that result when RPC is supplemented in periparturient cows could be indicative of a change in PC source.

Methionine is an AA that is used not only for protein synthesis, but also for synthesis of *S*-adenosylmethionine (**SAM**), one of the most important methyl donors in the body (Chiang et al., 1996). *S*-adenosylmethionine is required for de novo synthesis of PC via the PEMT pathway (Li and Vance, 2008), which is why Met has been investigated for its potential role as a lipotrope. Additionally, SAM is also implicated in the regulation of gene expression via histone methylation reactions (Mentch and Locasale, 2016).

Choline and Met metabolism are integrated through their participation in one-carbon metabolism. Choline indirectly serves as a methyl donor in the one-carbon metabolic pathway via betaine, the product of choline oxidation. Betaine is required for the regeneration of Met from homocysteine (**Hcy**) via the betaine-homocysteine S-methyltransferase (BHMT) pathway (Martinov et al., 2010). Alternatively, Hcy can also be converted to Met via the methyltetrahydrofolate pathway by methionine synthase, an enzyme that requires vitamin B_{12} to function (Preynat et al., 2009). Methionine also has the potential to indirectly influence PC synthesis via the PEMT pathway through SAM.

In an effort to derive nutritional strategies that improve cow health and performance during the periparturient period, attention has been given to nutrients involved in one-carbon metabolism such as choline and Met (Zhou et al., 2016b; Duplessis et al., 2017; Zenobi et al., 2018; Zang et al., 2019) due to their functions associated with lipid transport and metabolism, immunity, and gene expression. Previous investigations have shown that supplementing many of these nutrients during periods of negative energy balance that occur around the time of calving can modify hepatic expression of genes associated with choline, Met, and lipid metabolism (Goselink et al., 2013; Zhou et al., 2017a; Coleman et al., 2019a,b) as well as the plasma AA profile (Zhou et al., 2017b; Zang et al., 2019). However, only the study by Zhou et al. (2017a,b) investigated specific effects associated with supplementation of choline, Met, or both during the periparturient period. Recent attention has also been given to the effects of methyl donors on the blood choline metabolite profile in lactating cows (Wang et al., 2021) in an effort to elucidate mechanisms by which these nutrients affect metabolism and performance of postpartum cows. We hypothesized that choline and Met supplementation would modulate methyl group metabolism via (1) the CDP-choline pathway using choline derived from the diet; and (2) the PEMT pathway.

The objective of this study was to determine if providing supplemental choline, Met, or both during the periparturient period modifies plasma AA concentrations, the choline metabolic profile of plasma and milk, as well as hepatic expression of genes associated with choline, Met, and lipid metabolism.

MATERIALS AND METHODS

All procedures that utilized animals were approved by the University of Maryland, College Park, Institutional Animal Care and Use Committee (Protocol #857801–6).

Animals and Study Design

Between March and December of 2017, 25 primiparous and 29 multiparous Holstein cows from the Central Maryland Research and Education Center (Clarksville, MD) were utilized in a randomized block design experiment with a 2 × 2 factorial treatment structure. The 2 factors were 0 or 60 g/d RPC (ReaShure, Balchem Corporation) and 0 or 12 g/d RPM (Smartamine M; Adisseo USA, Inc.) prepartum and 18 g/d RPM postpartum. Thus, the 4 resulting treatments included: (1) no RPC or RPM (control; **CON**); (2) 60 g/d RPC (**CHO**); (3) 12 g/d RPM prepartum and 18 g/d RPM postpartum (**MET**); and (4) a combination of CHO and MET treatments (**CHO + MET**; 60 g/d RPC + 12 g/d RPM prepartum and 18 g/d RPM postpartum). Although not reported in our previous paper on production responses (Potts et al., 2020), the number of animals included in the study were based on a power analysis to provide an 80% chance of detecting a 2 kg/d difference in milk production across the main effects of CHO and MET within parity at $P \leq 0.05$. Before the start of the experiment, cows were blocked by age (primiparous vs. multiparous) and expected calving

date and randomly assigned to treatment. Treatments were applied daily as a top dress from 21 d before expected calving through 35 DIM. Because treatments were administered daily by project personnel, treatment assignments were not blinded. Additional details regarding basal pre- and postpartum diets and animal care and housing were described by Potts et al. (2020).

Sample Collection and Analysis

Blood was collected via coccygeal venipuncture before feeding on approximately −21, 7, and 14 d relative to calving. Because some cows calved early or late, prepartum samples were actually collected on −19 ± 2 d relative to calving. Samples were collected into one 10-mL evacuated tube containing potassium EDTA and one 10-mL evacuated tube containing sodium heparin. Tubes were immediately placed on ice and then centrifuged at 2,000 × *g* for 15 min at 4°C. Plasma aliquots were stored at −80°C until analysis.

Choline metabolite concentrations of plasma and milk (7 and 14 d postpartum) were determined by hydrophilic interaction liquid chromatography tandem MS (Artegoitia et al., 2014). Metabolites examined included betaine, free choline, glycerophosphocholine (**GPC**), sphingomyelin (**SM**), 16 species of PC, and 4 species of lysophosphatidylcholine (**LPC**). Amino acid concentrations of plasma were determined using the methods described by Sunny and Bequette (2010). Briefly, plasma samples (50 μL) were deproteinized with 500 μL of cold acetonitrile and evaporated to dryness under N_2. Amino acids and organic acids were then converted to their respective t-butyldimethylsilyl derivatives by heating at 90°C for 1 h. Metabolites were separated by GC (HP-5ms, 30 m × 0.25 mm × 0.25 μm, Agilent Technologies) before selected ion monitoring of specific ion fragments with MS under electron ionization.

Approximately 150 mg of liver tissue was collected from each animal via percutaneous liver biopsies at approximately −21 d (actual prepartum samples were obtained on −19 ± 2 d) and 7 d relative to calving using a 14-gauge biopsy needle (14-gauge × 15 cm; Tru-Cut, Merit Medical). Further details regarding the liver biopsy procedure are described by Potts et al. (2020). Tissue was immediately snap frozen in liquid nitrogen and stored at −80°C. Because 2 multiparous cows in block 6 and 1 multiparous cow in block 3 were removed early from the experiment due to health problems (Potts et al., 2020), gene expression analysis was completed on liver samples obtained from the 5 completed blocks of multiparous cows (n = 20, 5 cows per treatment).

Hepatic mRNA levels of genes related to choline and Met metabolism (*BHMT*; *PEMT*; phosphate cytidylyltransferase 1A, *PCYT1A*; glutathione synthetase, *GSS*; 5-methyltetrahydrofolate-homocysteine methyltransferase, *MTR*) and lipid metabolism (carnitine palmitoyltransferase 1A; diacylglycerol O-acyltransferase 1, *DGAT1*; 3-hydroxy-3-methylglutaryl-CoA synthase 2, *HMGCS2*; microsomal triglyceride transfer protein, *MTTP*; peroxisome proliferator activated receptor α, *PPARα*) were determined via real-time quantitative PCR. Primers were designed (Table 1) using the National Center for Biotechnology Information primer BLAST Software (2012). Primers needed to (1) span an intron, (2) target a region as close to the 3′ end of the sequence as possible, (3) amplify all splice variants, (4) have an annealing temperature of 58 to 60°C, (5) have a guanine:cytosine ratio of 40 to 60%, (6) be between 18 and 30 nucleotides in length, (6) generate a PCR product that was 100 to 250 nucleotides in length, and (7) specifically target the gene of interest. Primers (25 nmol DNA oligos; standard desalting purification) were obtained from Integrated DNA Technologies and reconstituted in ultrapure water. Amplification efficiencies were determined for each primer pair by performing reverse-transcription quantitative PCR (QuantiTect SYBR Green PCR Kit, Qiagen Inc.) using 2-fold serial dilutions of pooled cDNA (1 μg). Efficiency was calculated using the equation $[(10^{(-1/\text{slope})}) - 1]$, where slope is equal to the slope of the regression of the cycle threshold value on the $\log_{10}$ (copy number). Efficiencies ranged from 0.96 to 1.08 for all primers tested. Primer specificity was verified by dissociation curve analysis, agarose gel electrophoresis, and sequencing of PCR products.

Liver total RNA was extracted using the RNeasy Lipid Tissue Mini Kit with on-column DNase digestion (Qiagen Inc.). Approximately 20 to 30 mg of liver tissue was weighed and kept frozen in liquid nitrogen until homogenization in 0.5 mL of QIAzol Lysis Reagent (Qiagen Inc.). The remainder of the RNA extraction protocol was carried out according to manufacturer instructions. After extraction, RNA was stored at −80°C. The concentration of RNA in each sample was determined using a commercially available kit (Quant-iT RiboGreen RNA Assay Kit, Catalog #R11490, ThermoFisher Scientific) and 1 μg of RNA was used for cDNA synthesis (QuantiTect Reverse Transcription Kit; Qiagen Inc.). For the reverse transcription reactions, a reaction of a pool of total RNA without reverse transcriptase was conducted as a control for genomic DNA contamination. Complementary DNA was not diluted before PCR analysis and was stored at −20°C.

The PCR reactions were carried out using a commercially available kit according to the manufacturer instructions (QuantiTect SYBR Green PCR Kit; Qiagen Inc.). The PCR reactions were performed in 96-well plates (VWR International, LLC) in a CFX Connect Real-Time PCR Detection System (Bio-Rad Labora-

Table 1. Forward and reverse primers used for real-time quantitative PCR

Gene symbol	Description	GenBank accession no.	Primer sequence (5′ to 3′)[1]	Length[2]
Choline and methionine metabolism				
BHMT	Betaine-homocysteine S-methyltransferase	NM_001011679.1	F: GCTGTATGGGCAGTTGAAGC R: AATGGCAGTTTACCCCCACG	70
GSS	Glutathione synthetase	NM_001015630.1	F: GAAGATCGAACCCGAGCCTT R: GGTTCGAAGCAGATGTCCCA	157
MTR	5-methyltetrahydrofolate-homocysteine methyltransferase	NM_001030298.1	F: AGCCTTTCTTCGACGTCTGG R: CGTAGAAGGTGGCTATGGGC	264
PCYT1A	Phosphate cytidylyltransferase 1, choline, α	NM_001105052.2	F: GCTCCAACACAGAGGACAGAA R: ACTCTGCCTTGGACTGATGG	384
PEMT	Phosphatidylethanolamine N-methyltransferase	NM_182989.3	F: TCCGAGCGGGCATGACAAC R: CAAAGTGGGGTTCCGAGAGGT	197
Lipid metabolism				
CPT1A	Carnitine palmitoyltransferase 1A	NM_001304989.1	F: TCGTCACCATGCGTTACTCC R: GGGTTTTCGGCCTGAGAAGA	131
DGAT1	Diacylglycerol O-acyltransferase 1	NM_174693	F: TGGCCTTTCTCCTCGAGTCT R: TAGGTCAGGTTGTCGGGGTA	214
HMGCS2	3-hydroxy-3-methylglutaryl-CoA synthase 2	NM_001045883.1	F: GCAACACTGACATTGAGGGC R: ACAGACCACCAGTGCATAGC	125
MTTP	Microsomal triglyceride transfer protein	NM_001101834.1	F: GCTTCAGCTGCAATCTGGAC R: CACCGTGATGCCACCAGTTA	157
PPARα	Peroxisome proliferator activated receptor α	NM_001034036.1	F: CCGAGGAGTCATCCAGCATC R: TCAGCCGAATCGTTCTCCTAAA	129
Housekeeping				
PGK1	Phosphoglycerate kinase 1	NM_001034299.1	F: GAATGGGAAGCTTTTGCCCG R: AGCTCTAAACTGGCACCACC	176

[1]Primer direction: F = forward; R = reverse.

[2]Amplicon length in base pairs.

tories, Inc.) for 40 cycles using the following program: 95°C for 15 min (activation), 94°C for 15s (denaturation), 60°C for 30s (annealing), and 72°C for 30s (extension). Dissociation curve analysis ensured amplification of a single PCR product and absence from the no reverse-transcription and water controls. Data were normalized to phosphoglycerate kinase 1 and analyzed by the $2^{-\Delta\Delta Ct}$ method.

Statistical Analysis

Because production responses to treatment appeared to be different between primi- and multiparous cows (Potts et al., 2020), AA and choline metabolitic data were analyzed for primi- and multiparous cows separately. Postpartum blood AA and choline metabolite concentrations were analyzed using a repeated-measures mixed model (SAS Institute, version 9.4) that included the random effect of cow nested within block and fixed effects of wk relative to calving (1 or 2), the main effects of CHO and MET, and all 2- and 3-way interactions. Prepartum metabolite concentration, determined from samples collected in the morning before first treatment application (−19 ± 2 d relative to calving), was also included in the model as a covariate. Week relative to calving served as the repeated factor and a total of 8 covariance structures were tested, and the one that resulted in the lowest Akaike information criterion was selected for each variable. Milk choline metabolite yields were analyzed using a similar model that did not include a prepartum covariate measurement.

Hepatic mRNA level fold-changes (relative to postpartum CON) were analyzed in a mixed model that included the main effects of CHO and MET, their interaction, and the random effect of cow. Prepartum gene expression fold-change (relative to postpartum CON) was included in the model as a covariate. Thus, postpartum gene expression results are expressed as covariate-adjusted fold-changes relative to CON. Cook's distance was used to detect potential outliers using a threshold of 4 divided by number of observations ($4/n$). If a cow was identified as a potential outlier for more than 4 of the 10 genes of interest, she was removed from the analysis. Based on this method, results from 2 cows (one from CON and one from CHO) were considered outliers and removed from the final analysis. Statistical significance was declared at $P \leq 0.05$, and tendencies were considered at $P < 0.10$.

RESULTS

Milk Choline Metabolites

Multiparous Cows. Milk secretion of free choline, GPC, SM, and total PC was unaffected by CHO and MET (Table 2). However, CHO decreased betaine secretion during wk 1 postpartum (4,940 vs. 2,830 μmol/d) but not wk 2 (CHO × wk: $P = 0.04$). In contrast, feeding CHO increased total LPC secretion ($P = 0.05$). Aligning with this observation, CHO increased secretion of LPC 18:0 ($P < 0.01$), tended to increase secretion of LPC 16:0 ($P = 0.08$), and increased secretion of LPC 18:1 when fed without MET (CHO × MET: $P = 0.03$; Table 3).

Although secretion of total PC was unaffected, secretion of several individual PC species were elevated by CHO or MET. Feeding CHO increased secretion of PC 16:0/20:5 + 16:1/20:4, PC 18:1/22:6, and PC 18:0/22:6 + 18:1/22:5 and tended to increase secretion of PC 16:0/18:1 ($P < 0.01$, $P = 0.03$, $P = 0.02$, and $P = 0.08$, respectively; Table 3). Furthermore, CHO increased secretion of PC 18:1/20:4 + 18:0/20:5 + 16:0/22:5, PC 18:0/20:4, and PC 18:0/22:5 during wk 2 but not wk 1 postpartum (CHO × wk: $P = 0.05$, $P = 0.01$, and $P = 0.05$, respectively). Feeding MET increased secretion of PC 18:1/20:4 + 18:0/20:5 + 16:0/22:5, PC 18:0/20:4, PC 18:0/20:3, and PC 18:0/22:5 during wk 1 postpartum but not wk 2 (MET × wk: $P = 0.03$, $P < 0.01$, $P = 0.03$, and $P < 0.01$, respectively). In addition, feeding CHO without MET increased secretion of PC 18:0/22:6 + 18:1/22:5 and PC 18:0/18:1 during wk 2 postpartum but not wk 1, whereas MET increased their secretion irrespective of wk or CHO (CHO × MET × wk: both $P = 0.03$). In contrast, feeding CHO with MET increased secretion of PC 18:0/18:1 during wk 1 but not wk 2 postpartum, but CHO fed without MET increased its secretion during wk 2 postpartum (CHO × MET × wk: $P = 0.04$).

Primiparous Cows. Milk secretion of GPC, SM, and total PC was unaffected by CHO or MET. Feeding CHO tended to increase milk secretion of betaine ($P = 0.09$) and free choline ($P = 0.06$; Table 4). However, the effect of CHO on total LPC secretion was dependent on MET (CHO × MET: $P = 0.01$) as CHO increased total LPC secretion when fed without MET. Aligning with this result, CHO increased secretion of LPC 18:0 and LPC 16:0 (both $P = 0.02$; Table 5) when fed without MET. However, CHO tended to increase secretion of LPC 18:1 irrespective of MET ($P = 0.08$). In contrast, feeding MET with CHO suppressed the increase in secretion of LPC 18:2 during wk 1 but not wk 2 postpartum (CHO × MET × wk interaction: $P = 0.05$). The lack of CHO or MET effects on total PC secretion largely carried through to individual PC species (Table 5). However, CHO reduced secretion of PC 16:0/16:1 ($P = 0.04$) but increased secretion of PC 16:0/20:5 + 16:1/20:4 during wk 1 postpartum but not wk 2 (CHO × wk: $P = 0.03$). Additionally, MET increased secretion of PC 18:0/22:6 + 18:1/22:5

Table 2. Secretion rates (μmol/d) of major choline metabolites in milk for multiparous cows (n = 27) fed the control (CON) diet or the CON diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					Week postpartum			*P*-value			
Item[2]	CON	CHO	MET	CHO + MET	SEM	1	2	SEM	CHO	MET	MET × CHO	Week
Betaine[3]	3,669	2,773	3,933	2,902	605	3,885	2,754	269	0.07	0.62	0.86	0.04
Free choline	2,735	1,919	1,272	1,052	1,342	1,391	2,098	663	0.66	0.36	0.80	0.28
GPC	72,933	68,628	75,813	91,279	11,016	75,294	79,032	6,724	0.54	0.10	0.18	0.68
SM	1,335	1,240	1,239	1,167	157	1,466	1,024	95.8	0.52	0.41	0.91	<0.01
Total LPC	100	122	96	122	13.3	98.0	122	8.14	0.05	0.80	0.77	0.05
Total PC	3,827	4,721	3,828	4,733	754	4,314	4,240	461	0.17	0.99	0.99	0.90

[1]Prepartum, cows were supplemented with 60 g/d RPC (ReaShure, Balchem Corp.; CHO), 12 g/d RPM (Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Milk samples were obtained during wk 1 and 2 postpartum.
[2]GPC = glycerophosphocholine; SM = sphingomyelin; LPC = lysophosphatidylcholine; PC = phosphatidylcholine.
[3]CHO × week interaction, $P \leq 0.05$.

($P = 0.05$) and tended to increase secretion of PC 18:0/22:5 ($P = 0.06$).

Blood Choline Metabolites

Multiparous Cows. There were no effects of CHO or MET on plasma betaine, free choline, GPC, or total PC concentrations (Table 6). However, CHO decreased plasma SM concentrations ($P = 0.04$) and MET tended to decrease total LPC concentrations ($P = 0.09$). In line with the latter observation, MET reduced plasma concentrations of LPC 18:0 ($P = 0.05$; Table 5). In addition, feeding CHO increased plasma concentrations of LPC 16:0 and LPC 18:1, but this effect was dependent upon MET (CHO × MET: $P = 0.05$ and $P = 0.02$, respectively). Plasma total PC concentrations were unaffected and individual PC species were largely unaffected by CHO or MET (Table 7). However, CHO increased concentration of PC 18:0/18:1 during wk 1 postpartum (CHO × wk: $P = 0.05$) and tended to increase plasma concentrations of PC 16:0/20:4 when fed without MET (CHO × MET: $P = 0.09$). Furthermore, feeding CHO and MET together tended to reduce PC 16:0/18:2 concentrations (CHO × MET: $P = 0.09$).

Primiparous Cows. Plasma concentrations of betaine, GPC, and SM were unaffected by CHO or MET but MET tended to increase plasma free choline ($P = 0.09$; Table 8). Total LPC and individual LPC species concentrations were also unaffected by CHO or MET (Tables 8 and 9, respectively). There was no effect of CHO on total PC concentrations and the effect on individual PC species was minimal as only PC 18:0/18:1 tended to be reduced by CHO ($P = 0.06$). Feeding MET decreased total PC concentrations at wk 2 postpartum (MET × wk: $P = 0.01$; Table 9). Aligning with this observation, MET reduced concentrations of several PUFA-PC species (including PC 16:0/20:4, PC 18:1/20:4 + 18:0/20:5 + 16:0/22:5, PC 18:0/20:4, PC 18:0/20:3, and PC 18:1/22:6) during wk 2 postpartum (MET × wk: $P = 0.02$, $P = 0.02$, $P = 0.04$, $P = 0.04$, and $P = 0.05$, respectively). Similarly, MET decreased concentrations of all SFA- and UFA-PC species during wk 2 postpartum (MET × wk: all $P < 0.01$; Table 9). In contrast, MET fed without CHO increased concentrations of PC 16:0/20:3 during wk 1 postpartum, whereas other treatments failed to attain similar levels until wk 2 postpartum (CHO × MET × wk: $P = 0.03$).

Blood Amino Acids

Multiparous Cows. Feeding MET increased plasma Met concentrations ($P < 0.01$) but reduced Ser concentrations during wk 2 postpartum (MET × wk: $P = 0.03$; Table 10). Feeding MET without CHO

Table 3. Secretion rates (μmol/d) of lysophosphatidylcholine (LPC) and phosphatidylcholine (PC) species in milk for multiparous cows (n = 27) fed the control (CON) diet or the CON diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					Week postpartum			*P*-value			
Item	CON	CHO	MET	CHO + MET	SEM	1	2	SEM	CHO	MET	MET × CHO	Week
LPC species												
LPC 16:0	79.2	90.1	70.6	93.7	11.0	75.0	91.8	6.74	0.08	0.73	0.40	0.09
LPC 18:2	10.0	12.3	10.8	8.85	2.44	8.9	12.0	1.49	0.92	0.42	0.19	0.14
LPC 18:1	1.73[ab]	2.85[a]	1.72[b]	1.34[b]	0.46	1.86	1.96	0.28	0.33	0.03	0.03	0.80
LPC 18:0	10.6	16.3	12.7	17.0	1.41	13.4	14.9	0.86	<0.01	0.14	0.43	0.22
Long-chain/PUFA PC species												
PC 16:0/20:5, 16:1/20:4	0.08	0.38	0.17	0.27	0.08	0.16	0.05	0.05	<0.01	0.88	0.08	0.06
PC 16:0/20:4	140	180	149	133	29.6	154	147	18.0	0.62	0.33	0.15	0.80
PC 16:0/20:3	322	442	374	393	67.7	385	380	41.3	0.23	0.97	0.25	0.93
PC 16:0/22:6, 18:1/22:5, 18:2/20:4	11.3	21.5	21.0	21.4	3.75	19.0	18.5	2.29	0.11	0.07	0.06	0.88
PC 18:1/20:4, 18:0/20:5, 16:0/22:5[2,3]	38.3	61.2	56.0	62.2	8.74	55.1	53.7	5.34	0.06	0.12	0.15	0.84
PC 18:0/20:4[2,3]	23.4	40.0	39.7	46.8	5.59	37.1	37.9	3.42	0.02	<0.01	0.20	0.86
PC 18:0/20:3[3]	9.2	15.1	14.9	16.2	2.49	14.6	13.1	1.52	0.10	0.05	0.16	0.47
PC 18:1/22:6[4]	0.99	2.80	3.97	4.98	0.72	2.74	3.63	0.44	0.03	<0.01	0.39	0.15
PC 18:0/22:6, 18:1/22:5[4]	2.07	6.25	5.87	8.18	1.48	5.04	6.15	0.90	0.02	0.01	0.33	0.37
PC 18:0/22:5[2,3]	2.30	5.09	5.76	7.13	1.08	5.50	4.64	0.66	0.04	<0.01	0.31	0.34
Saturated/unsaturated PC species												
PC 16:0/16:1	159	151	130	163	35.0	143	159	21.4	0.66	0.72	0.37	0.58
PC 16:0/16:0	521	615	500	596	112.2	527	589	68.5	0.31	0.78	0.99	0.50
PC 16:0/18:2	451	560	433	441	89	509	433	54.4	0.43	0.25	0.38	0.31
PC 16:0/18:1	1,176	1,677	1,193	1,461	245	1,396	1,358	149	0.08	0.54	0.46	0.85
PC 18:0/18:2, 18:1/18:1	687	895	688	854	141.5	805	757	86.4	0.13	0.83	0.82	0.68
PC 18:0/18:1[4]	219	324	281	383	44.5	306	298	27.2	0.02	0.06	0.96	0.83

[a,b]Means with differing superscripts are significantly different ($P < 0.05$).

[1]Prepartum, cows were supplemented with 60 g/d RPC (ReaShure, Balchem Corp.; CHO), 12 g/d RPM (Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Milk samples were obtained during wk 1 and 2 postpartum.

[2]CHO × week interaction, $P \leq 0.05$.

[3]MET × week interaction, $P \leq 0.05$.

[4]MET × CHO × week interaction, $P \leq 0.05$.

Table 4. Secretion rates (μmol/d) of major choline metabolites in milk for primiparous cows (n = 24) fed the control (CON) diet or the CON diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					Week postpartum			*P*-value			
Item[2]	CON	CHO	MET	CHO + MET	SEM	1	2	SEM	CHO	MET	MET × CHO	Week
Betaine	2,412	3,215	2,534	2,845	319	3,317	2,185	181	0.09	0.69	0.44	<0.01
Free choline	962	1,242	1,049	1,162	100	768	1,439	68.5	0.06	0.97	0.39	<0.01
GPC	44,534	44,371	43,049	49,422	4,399	42,865	47,823	2,557	0.49	0.69	0.47	0.06
SM	765	915	855	877	62.5	840	865	37.4	0.17	0.67	0.31	0.55
Total LPC	66.6[b]	103[a]	84.1[ab]	73.1[b]	8.58	72.9	90.6	5.07	0.14	0.47	0.01	<0.01
Total PC	2,608	2,408	2,966	2,367	314	2,641	2,532	197	0.20	0.61	0.52	0.65

[a,b]Means with differing superscripts are significantly different ($P < 0.05$).

[1]Prepartum, cows were supplemented with 60 g/d RPC (ReaShure, Balchem Corp.; CHO), 12 g/d RPM (Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Milk samples were obtained during wk 1 and 2 postpartum.

[2]GPC = glycerophosphocholine; SM = sphingomyelin; LPC = lysophosphatidylcholine; PC = phosphatidylcholine.

increased Phe concentrations and tended to increase Ile, Leu, and total branched-chain AA concentrations (CHO × MET: $P = 0.04$, $P = 0.06$, $P = 0.09$, and $P = 0.08$, respecitvely). There were few effects of CHO on plasma AA, but feeding CHO without MET decreased glutamate concentrations during wk 1 but not wk 2 postpartum (CHO × MET × wk: $P = 0.03$).

Primiparous Cows. Results for plasma AA in primiparous cows are shown in Table 11. Feeding MET increased plasma Met concentrations ($P < 0.01$) but tended to reduce Gly concentrations ($P = 0.08$). Feeding MET reduced Ser concentrations ($P = 0.01$), but this effect tended to be intensified when CHO was fed with MET (CHO × MET: $P = 0.08$). The effect of CHO on plasma AA were minimal, although CHO increased Gln and tended to increase Gly concentrations during wk 1 postpartum (CHO × wk: $P = 0.04$ and $P = 0.07$, respectively).

Hepatic Gene Expression

Relative mRNA levels of hepatic genes associated with choline, Met, and fatty acid metabolism for a subset of multiparous cows (n = 20) is shown in Table 12. Of the one-carbon metabolism genes examined, only *BHMT* and *PCYT1A* were affected by CHO or MET. Feeding MET reduced mRNA levels of *PCYT1A* ($P < 0.01$). Additionally, CHO tended to increase mRNA levels of *BHMT* and *PCYT1A* when fed without MET (Figure 1, CHO × MET: $P = 0.09$ and $P = 0.10$, respectively). The mRNA levels of *HMGCS2* and *PPARα* tended to be reduced by CHO ($P = 0.10$ and $P = 0.08$). Feeding MET tended to reduce mRNA expression of *DGAT1* ($P = 0.08$).

DISCUSSION

Our results suggest that supplemental choline and Met both seem to affect one-carbon metabolic pathways during the periparturient period but that responses are inconsistent between primi- and multiparous cows and appear to be dependent on Met availability. Although we did not examine metabolite flux through pathways associated with choline and Met metabolism, changes in metabolite secretion in milk and concentrations in plasma are indicative of shifts in these pathways. Alterations in hepatic expression of genes associated with one-carbon and lipid metabolism in multiparous cows further support this idea. However, the tendency for interactive effects on hepatic gene expression suggest that supplemental choline may play a role in promoting the CDP-choline and BHMT pathways but that the response is dependent on Met availability. Interactive

Table 5. Secretion rates (μmol/d) of lysophosphatidylcholine (LPC) and phosphatidylcholine (PC) species in milk for primiparous cows (n = 24) fed the control (CON) diet or the CON diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					Week postpartum			*P*-value			
Item	CON	CHO	MET	CHO + MET	SEM	1	2	SEM	CHO	MET	MET × CHO	Week
LPC species												
LPC 16:0	54.1[b]	83.5[a]	67.1[ab]	57.6[b]	7.41	58.8	72.3	4.3	0.18	0.39	0.02	<0.01
LPC 18:2[2]	4.27	5.31	4.88	5.39	0.59	4.07	5.86	0.32	0.19	0.55	0.65	<0.01
LPC 18:1	0.93	1.07	0.97	1.33	0.14	0.06	1.19	0.08	0.08	0.28	0.42	<0.01
LPC 18:0	8.3[b]	12.9[a]	10.7[ab]	8.4[b]	1.33	9.3	10.8	0.75	0.41	0.41	0.02	0.08
Long-chain/PUFA PC species												
PC 16:0/20:5, 16:1/20:4[3]	0.05	0.06	0.13	0.25	0.05	0.20	0.05	0.04	0.17	<0.01	0.20	<0.01
PC 16:0/20:4	71.1	68.6	81.0	65.7	10.4	73.7	69.6	6.44	0.39	0.74	0.53	0.60
PC 16:0/20:3	191	212	250	197	28.4	221	204	18.1	0.56	0.45	0.20	0.48
PC 16:0/22:6, 18:1/22:5, 18:2/20:4	7.6	9.0	12.7	9.61	1.68	10.9	8.5	0.95	0.62	0.10	0.18	0.02
PC 18:1/20:4, 18:0/20:5, 16:0/22:5	29.9	26.2	37.5	30.5	4.45	35.4	26.7	2.55	0.23	0.19	0.69	<0.01
PC 18:0/20:4	18.1	18.2	24.9	19.3	2.93	22.9	17.3	1.77	0.35	0.19	0.32	0.01
PC 18:0/20:3	6.19	6.43	9.02	7.11	1.10	8.06	6.32	0.63	0.44	0.12	0.32	0.02
PC 18:1/22:6	1.81	2.23	3.42	2.47	0.60	2.93	2.04	0.35	0.65	0.13	0.25	0.03
PC 18:0/22:6, 18:1/22:5	3.07	3.66	5.59	4.22	0.74	4.78	3.49	0.45	0.59	0.05	0.19	0.02
PC 18:0/22:5	2.48	2.42	4.02	3.37	0.62	3.81	2.33	0.38	0.56	0.06	0.63	<0.01
Saturated/unsaturated PC species												
PC 16:0/16:1	114	96	119	88.1	11.1	96.5	112	8.5	0.04	0.91	0.56	0.28
PC 16:0/16:0	341	324	380	296	47.8	323	347	32.0	0.29	0.91	0.48	0.57
PC 16:0/18:2	256	239	261	210	33.2	244	239	20.9	0.30	0.73	0.60	0.85
PC 16:0/18:1	793	796	930	757	107	835	803	67.0	0.42	0.64	0.41	0.70
PC 18:0/18:2, 18:1/18:1	482	478	608	479	59.4	553	471	41.7	0.30	0.33	0.33	0.14
PC 18:0/18:1	201	182	231	174	27.0	215	179	15.5	0.16	0.67	0.47	0.04

[a,b]Means with differing superscripts are significantly different ($P < 0.05$).

[1]Prepartum, cows were supplemented with 60 g/d RPC (ReaShure, Balchem Corp.; CHO), 12 g/d RPM (Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Milk samples were obtained during wk 1 and 2 postpartum.

[2]MET × CHO × week interaction, $P \leq 0.05$.

[3]CHO × week interaction, $P \leq 0.05$.

Table 6. Postpartum plasma choline metabolite concentrations (μ*M*) for multiparous cows (n = 27) fed the control (CON) diet or the CON diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					Week postpartum			*P*-value			
Item[2]	CON	CHO	MET	CHO + MET	SEM	1	2	SEM	CHO	MET	MET × CHO	Week
Betaine	49.9	42.6	42.7	41.1	7.56	48.5	39.7	3.16	0.58	0.49	0.64	<0.01
Free choline	4.21	4.59	4.07	3.75	0.44	4.52	3.79	0.22	0.95	0.15	0.29	0.02
GPC	11.6	18.4	11.1	13.1	4.49	8.6	18.5	3.93	0.29	0.45	0.53	0.03
SM	338	259	340	250	37.9	275	319	18	0.04	0.92	0.85	0.01
Total LPC	250	296	238	202	35.2	197	296	30.2	0.88	0.09	0.16	0.01
Total PC	2,406	2,583	2,528	2,094	370	2,104	2,701	230	0.74	0.57	0.34	0.03

[1]Prepartum, cows were supplemented with 60 g/d RPC (ReaShure, Balchem Corp.; CHO), 12 g/d RPM (Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Blood samples were obtained during wk 1 and 2 postpartum.

[2]GPC = glycerophosphocholine; SM = sphingomyelin; LPC = lysophosphatidylcholine; PC = phosphatidylcholine.

effects on milk and plasma metabolites provide additional evidence of this.

Choline Metabolites

Baseline milk secretion and plasma concentrations of free choline, betaine, SM, total LPC, and total PC were comparable between our study and that of previous studies (Artegoitia et al., 2014; de Veth et al., 2016; Zenobi et al., 2018; Wang et al., 2021). Free choline and betaine concentrations in plasma appear to be the most responsive to postruminal choline supplied via abomasal infusion (de Veth et al., 2016), but postruminal choline supplied via RPC has not been shown to affect plasma betaine or free choline (de Veth et al., 2016; Zenobi et al., 2018; S. B. Potts, E. Davis, and R. A. Erdman, University of Maryland, College Park, MD, unpublished data). By contrast, de Veth et al. (2016) indicated an increase in milk free choline and betaine secretion when RPC was fed to lactating cows. In our study, CHO decreased milk secretion of betaine during the first wk postpartum in multiparous cows, which coincides with our observation for the tendency of CHO to increase hepatic BHMT expression for multiparous cows when fed without MET. This increased use of betaine by the liver likely reduced betaine availability to the mammary gland. In addition to putative increases in dietary choline oxidation to betaine, the overall increase in milk secretion of LPC and several LPC species for cows fed CHO also suggest that CHO increased choline metabolite production via the CDP-choline or *PEMT* pathways.

Feeding CHO tended to increase milk secretion of free choline and betaine for primiparous cows in our study, which is contrary to our observations for multiparous cows but similar to the findings reported by de Veth et al. (2016). This increase in free choline could suggest an adequate choline supply and increased ability to support the BHMT pathway for Met synthesis. However, in their study of choline and Met kinetics in lactating goats, Emmanuel and Kennelly (1984) indicated that very little choline is directed toward Met synthesis, which would explain why there was no effect of CHO on plasma Met concentrations in our study. Varied responses to CHO between primi- and multiparous cows for milk secretion of choline metabolites in our study are likely due, in part, to differences in physiological state and production level, which presumably affect the demand for methyl donors. Younger cows seem to have lower choline and Met requirements than older cows (Potts et al., 2020; Swartz et al., 2022) and methyl donor demands are presumably greater for high-producing, early-lactation cows (Pinotti et al., 2002). Therefore, it is possible that the methyl donor

Table 7. Postpartum plasma lysophosphatidylcholine (LPC) and phosphatidylcholine (PC) species concentrations (μM) for multiparous cows (n = 27) fed the control (CON) diet or the control diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					Week postpartum			*P*-value			
Item	CON	CHO	MET	CHO + MET	SEM	1	2	SEM	CHO	MET	MET × CHO	Week
LPC species												
LPC 16:0	70.5[ab]	80.0[a]	69.7[ab]	52.6[b]	7.59	60.1	76.3	5.59	0.61	0.04	0.05	0.02
LPC 18:2	125	153	122	107	20.5	96.2	158	17.5	0.74	0.16	0.21	<0.01
LPC 18:1	4.25[b]	6.28[a]	4.61[b]	4.30[b]	0.59	4.02	5.70	0.42	0.16	0.11	0.02	<0.01
LPC 18:0	43.4	58.1	39.2	39.3	6.25	39.5	50.5	4.32	0.26	0.05	0.15	0.08
Long-chain/PUFA PC species												
PC 16:0/20:5, 16:1/20:4	5.14	7.73	4.93	5.11	1.21	5.28	6.18	0.66	0.25	0.17	0.21	0.31
PC 16:0/20:4	101	129	108	89	16.0	98	115	8.91	0.78	0.23	0.09	0.15
PC 16:0/20:3	188	216	227	190	25.9	181	230	14.2	0.89	0.76	0.14	0.02
PC 16:0/22:6, 18:1/22:5, 18:2/20:4	11.7	14.6	12.2	9.8	2.42	11.7	12.5	1.35	0.92	0.29	0.18	0.66
PC 18:1/20:4, 18:0/20:5, 16:0/22:5	71.9	86.1	74.6	59.3	14.94	66.2	79.7	8.10	0.97	0.33	0.22	0.22
PC 18:0/20:4	129	171	148	139	36.0	126	168	19.6	0.64	0.82	0.38	0.12
PC 18:0/20:3	94.2	115	121	125	30.1	91.3	136	16.8	0.70	0.45	0.72	0.06
PC 18:1/22:6	2.99	4.51	3.15	3.02	0.62	3.22	3.61	0.35	0.28	0.20	0.10	0.41
PC 18:0/22:6, 18:1/22:5	10.9	14.9	10.0	8.8	2.95	10.4	12.0	1.62	0.63	0.16	0.27	0.47
PC 18:0/22:5	41.7	56.9	42.9	36.6	11.5	37.4	51.6	6.15	0.69	0.31	0.25	0.09
Saturated/unsaturated PC species												
PC 16:0/16:1	16.7	16.3	16.9	14.9	3.2	14.6	17.9	1.62	0.71	0.83	0.75	0.18
PC 16:0/16:0	13.8	17.6	14.9	12.4	2.9	12.9	16.4	1.46	0.83	0.40	0.18	0.11
PC 16:0/18:2	608	678	674	511	79	547	688	39.5	0.55	0.44	0.09	0.01
PC 16:0/18:1	286	293	266	228	33	254	283	17.4	0.64	0.15	0.40	0.19
PC 18:0/18:2, 18:1/18:1	586	720	681	628	73.0	578	729	42.2	0.59	0.98	0.13	0.02
PC 18:0/18:1[2]	216	244	207	216	28.0	201	241	15.8	0.52	0.43	0.67	0.07

[a,b]Means with differing superscripts are significantly different ($P < 0.05$).

[1]Prepartum, cows were supplemented with 60 g/d rumen-protected choline (RPC; ReaShure, Balchem Corp.; CHO), 12 g/d rumen-protected Met (RPM; Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Blood samples were obtained during wk 1 and 2 postpartum.

[2]CHO × week interaction ($P \leq 0.05$).

Table 8. Postpartum plasma choline metabolite concentrations (μ*M*) for primiparous cows (n = 24) fed the control (CON) diet or the CON diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					Week postpartum			*P*-value			
Item[2]	CON	CHO	MET	CHO + MET	SEM	1	2	SEM	CHO	MET	MET × CHO	Week
Betaine	52.7	51.2	45.8	48.4	6.53	52.8	46.3	3.27	0.93	0.42	0.74	0.03
Free choline	4.07	3.84	4.72	4.30	0.30	4.30	4.16	0.20	0.25	0.09	0.74	0.61
GPC	9.8	13.9	13.6	15.1	3.08	12.7	13.5	1.97	0.32	0.38	0.65	0.79
SM	299	288	311	314	27.5	262	344	18	0.87	0.46	0.80	<0.01
Total LPC	222	268	247	227	23.0	215	267	14.6	0.52	0.72	0.13	0.02
Total PC[3]	2,458	2,337	2,318	2,160	176	1,869	2,767	114	0.40	0.34	0.91	<0.01

[1]Prepartum, cows were supplemented with 60 g/d RPC (ReaShure, Balchem Corp.; CHO), 12 g/d RPM (Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Blood samples were obtained during wk 1 and 2 postpartum.

[2]GPC = glycerophosphocholine; SM = sphingomyelin; LPC = lysophosphatidylcholine; PC = phosphatidylcholine.

[3]MET × week interaction, $P \leq 0.05$.

balance of the primiparous cows in our study was more similar to that of the mid-lactation cows studied by de Veth et al., (2016), which may help explain the agreement of responses to CHO between the primiparous, but not multiparous, cows in our study and the cows in the study conducted by de Veth et al. (2016). Although CHO did not affect plasma betaine or free choline concentrations in primiparous cows, MET tended to increase plasma free choline concentrations, which suggests either a preferential increase in the use of Met as a methyl donor over choline or that Met was used to synthesize additional choline (Emmanuel and Kennelly, 1984). In contrast, Zhou et al. (2016a) reported no effect of MET on plasma free choline concentrations in multiparous, periparturient cows, which agrees with our observations for multiparous cows.

Plasma concentration and milk secretion of SM was unaffected by treatment in primiparous cows, which agrees with data reported by de Veth et al. (2016). Milk secretion of SM was also unaffected by treatment in multiparous cows. However, the decrease in plasma SM concentrations for multiparous cows fed CHO was somewhat surprising, given that previous work by Zenobi et al. (2018) indicated an increase in plasma SM when dry cows were fed RPC and de Veth et al. (2016) showed no effect of postruminal choline supply on milk or plasma SM. Sphingomyelin is produced from PC and is involved with cell signaling associated with inflammation (Nixon, 2009; Avota et al., 2019), so a decrease in SM could suggest a reduced inflammatory response. This presents a mechanism by which choline may exert favorable effects with regard to minimizing inflammation during the periparturient period. Sphingomyelin has also been shown to be inversely associated with insulin sensitivity in monogastrics (Chang et al., 2019; Li et al., 2011), which would suggest greater insulin sensitivity for multiparous cows fed CHO in our study. However, additional research is needed to determine what, if any, effects dietary choline has on insulin sensitivity in dairy cows.

Lysophosphatidylcholine and Phosphatidylcholine

Work presented by de Veth et al. (2016) indicated no change in total LPC secretion in milk when supplemental choline was delivered via abomasal infusion or as RPC. The increased milk secretion of LPC in our study combined with the trend for reduced betaine secretion, is supportive of increased use of choline for PC synthesis. Because GPC secretion was also unaffected, CHO may have diverted additional PC toward LPC production, rather than to GPC. Our observations for primiparous cows were similar, although this effect seemed to depend on Met availability. This was largely

Table 9. Postpartum plasma lysophosphatidylcholine (LPC) and phosphatidylcholine (PC) species concentrations (μM) for primiparous cows (n = 24) fed the control (CON) diet or the CON diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					Week postpartum			*P*-value			
Item	CON	CHO	MET	CHO + MET	SEM	1	2	SEM	CHO	MET	MET × CHO	Week
LPC species												
LPC 16:0	70.4	79.0	73.3	68.5	6.28	67.3	78.3	4.06	0.74	0.52	0.25	0.06
LPC 18:2	102	120	120	111	12.2	97.0	129	7.85	0.66	0.73	0.22	<0.01
LPC 18:1	3.87	4.79	4.29	4.10	0.46	3.62	4.91	0.30	0.39	0.76	0.19	<0.01
LPC 18:0	46.2	51.4	52.8	43.1	4.90	42.8	53.9	3.18	0.62	0.85	0.11	0.02
Long-chain/PUFA PC species												
PC 16:0/20:5, 16:1/20:4	4.93	5.09	5.37	3.51	0.85	4.04	5.41	0.54	0.27	0.51	0.19	0.08
PC 16:0/20:4[2]	92	100	90	85	9.4	74	110	6.06	0.89	0.35	0.46	<0.01
PC 16:0/20:3[3]	191	182	190	177	22.4	143	227	11.3	0.60	0.87	0.93	<0.01
PC 16:0/22:6, 18:1/22:5, 18:2/20:4	15.2	14.3	15.8	12.4	1.92	11.8	17.1	1.24	0.22	0.75	0.47	<0.01
PC 18:1/20:4, 18:0/20:5, 16:0/22:5[2]	79.5	86.3	70.1	64.7	8.74	59.2	91.1	5.63	0.93	0.08	0.45	<0.01
PC 18:0/20:4[2]	123	122	128	104	18.0	94.0	145	11.7	0.45	0.72	0.50	<0.01
PC 18:0/20:3[2]	92.7	88.9	89.8	78.2	13.4	65.6	109	8.7	0.54	0.60	0.76	<0.01
PC 18:1/22:6[2]	3.9	3.5	3.3	2.9	0.4	2.7	4.1	0.3	0.24	0.09	0.98	<0.01
PC 18:0/22:6, 18:1/22:5	18.6	19.0	18.6	12.7	2.54	13.7	20.7	1.64	0.25	0.19	0.19	<0.01
PC 18:0/22:5	56.9	57.9	53.2	34.3	8.74	39.3	61.8	5.48	0.25	0.12	0.22	<0.01
Saturated/unsaturated PC species												
PC 16:0/16:1[2]	15.2	19.4	16.8	15.1	2.8	13.7	19.6	1.41	0.64	0.60	0.26	<0.01
PC 16:0/16:0[2]	13.7	15.5	14.3	11.6	2.1	11.0	16.6	1.41	0.83	0.40	0.28	<0.01
PC 16:0/18:2[2]	535	543	519	494	60	429	616	30.0	0.88	0.57	0.77	<0.01
PC 16:0/18:1[2]	285	275	303	278	32	251	320	15.8	0.57	0.71	0.80	<0.01
PC 18:0/18:2, 18:1/18:1[2]	627	555	583	554	67.7	452	708	34.5	0.42	0.72	0.73	<0.01
PC 18:0/18:1[2]	276	227	259	239	19.7	210	290	12.6	0.06	0.89	0.45	<0.01

[1]Prepartum, cows were supplemented with 60 g/d RPC (ReaShure, Balchem Corp.; CHO), 12 g/d RPM (Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Blood samples were obtained during wk 1 and 2 postpartum.

[2]MET × week interaction, $P \leq 0.05$.

[3]MET × CHO × week interaction, $P \leq 0.05$.

Table 10. Postpartum plasma amino acid concentrations (μM) for multiparous cows (n = 27) fed the control (CON) diet or the CON diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					Week postpartum			*P*-value			
Item[2]	CON	CHO	MET	CHO + MET	SEM	1	2	SEM	CHO	MET	MET × CHO	Week
Alanine	241	188	214	207	25.1	195.4	229.1	10.2	0.25	0.83	0.33	<0.01
Aspartate	4.90	3.56	4.49	4.34	0.50	4.25	4.39	0.22	0.17	0.65	0.14	0.55
Glutamate[3]	35.1	29.4	36.4	34.8	3.16	32.99	34.88	1.4	0.26	0.21	0.42	0.24
Glutamine	57.3	53.1	50.0	61.9	15.1	53.4	57.8	6.25	0.80	0.95	0.49	0.45
Glycine	569	779	703	712	151	727	654	78.1	0.48	0.78	0.38	0.50
Histidine	58.8	67.7	66.3	71.0	6.54	64.4	67.5	2.85	0.31	0.31	0.67	0.27
Isoleucine	198	215	223	172	22.3	191.7	212.6	11.5	0.44	0.63	0.06	0.18
Leucine	187	183	214	162	17.9	176.2	196.3	9.57	0.12	0.82	0.09	0.12
Lysine	82.1	73.2	90.0	75.1	8.85	76.6	83.6	5.26	0.23	0.53	0.68	0.27
Methionine	22.8	22.6	34.3	35.9	3.48	28.9	28.9	1.47	0.84	<0.01	0.74	0.99
Phenylalanine	53.4[ab]	53.1[ab]	60.1[a]	47.8[b]	3.81	52.8	54.4	1.92	0.13	0.80	0.04	0.53
Proline	105	89.8	99.8	91.6	10.8	89.2	104	5.52	0.41	0.87	0.74	<0.01
Serine[4]	134	121	107	120	9.26	114	127	5.04	0.99	0.08	0.11	0.12
Threonine	105	85.9	98.8	88.7	17.0	88.9	100.1	6.12	0.40	0.90	0.73	0.02
Tyrosine	44.0	42.4	46.2	43.3	5.87	38.9	49.1	2.54	0.70	0.74	0.89	<0.01
Valine	257	248	301	236	23.7	247	274	13.7	0.19	0.95	0.13	0.02
BCAA	656	658	741	567	63.0	616	695	33.1	0.18	0.96	0.08	0.08
EAA	997	1010	1153	964	82.2	989	1074	50.4	0.33	0.43	0.15	0.16

[a,b]Means with differing superscripts are significantly different ($P < 0.05$).

[1]Prepartum, cows were supplemented with 60 g/d RPC (ReaShure, Balchem Corp.; CHO), 12 g/d RPM (Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Blood samples were obtained during wk 1 and 2 postpartum.

[2]BCAA = branched-chain AA; EAA = essential AA.

[3]MET × CHO × week interaction, $P \leq 0.05$.

[4]MET × week interaction, $P \leq 0.05$.

Table 11. Postpartum plasma amino acid concentrations (μM) for primiparous cows (n = 24) fed the control (CON) diet or the CON diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					Week postpartum			*P*-value			
Item[2]	CON	CHO	MET	CHO + MET	SEM	1	2	SEM	CHO	MET	MET × CHO	Week
Alanine	191	213	209	209	14.4	196	215	8.90	0.40	0.60	0.41	0.15
Aspartate	4.96	5.51	5.50	4.69	0.36	5.15	4.94	0.21	0.74	0.23	0.16	0.48
Glutamate	45.1	46.4	47.2	45.2	2.61	47	45	1.66	0.89	0.85	0.50	0.42
Glutamine[3]	72.4	78.1	77.3	87.3	7.5	81	77	4.71	0.25	0.32	0.75	0.50
Glycine	725	989	638	562	150	757	700	91.9	0.48	0.08	0.21	0.66
Histidine	64.9	67.5	63.7	64.8	3.30	64.5	65.9	2.05	0.54	0.51	0.81	0.62
Isoleucine	161	171	184	176	14.2	163	183	8.85	0.95	0.28	0.46	0.11
Leucine	158	165	170	159	15.0	155	171	10.1	0.89	0.81	0.54	0.10
Lysine	63.2	71.1	74.0	73.8	9.31	65.0	76.1	5.85	0.63	0.39	0.63	0.05
Methionine	22.2	24.1	36.2	35.3	2.21	29.4	29.5	1.15	0.80	<0.01	0.49	0.90
Phenylalanine	51.3	56.2	53.1	49.8	2.64	53.3	51.9	1.67	0.74	0.33	0.10	0.55
Proline	87.6	96.2	88.3	94.1	6.91	86.3	96.8	4.51	0.26	0.91	0.82	0.03
Serine	113	127	106	95	8.19	111	110	4.75	0.90	0.01	0.08	0.96
Threonine	76.2	77.7	70.2	70.2	6.9	64.8	82.3	4.65	0.91	0.29	0.91	<0.01
Tyrosine	41.6	41.6	40.2	34.9	3.76	35.9	43.6	2.8	0.45	0.25	0.44	0.01
Valine	225	219	231	216	15.5	205	241	9.61	0.47	0.91	0.73	0.01
BCAA	543	556	585	552	41.0	523	595	25.5	0.79	0.61	0.54	0.05
EAA	813	866	893	893	95.3	813	919	58.1	0.75	0.51	0.74	0.04

[1]Prepartum, cows were supplemented with 60 g/d RPC (ReaShure, Balchem Corp.; CHO), 12 g/d RPM (Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Blood samples were obtained during wk 1 and 2 postpartum.

[2]BCAA = branched-chain AA; EAA = essential AA.

[3]CHO × week interaction, $P \leq 0.05$.

Table 12. Postpartum covariate-adjusted gene expression fold-changes relative to control (CON) for multiparous cows (n = 20) fed the CON diet or the CON diet plus rumen-protected choline (RPC), rumen-protected Met (RPM), or both[1]

	Treatment					*P*-value		
Gene symbol[2]	CON	CHO	MET	CHO + MET	SEM	CHO	MET	MET × CHO
Choline and methionine metabolism								
BHMT	0.94	2.10	1.07	1.03	0.36	0.13	0.17	0.09
GSS	1.52	1.73	1.59	1.51	0.31	0.72	0.69	0.43
MTR	1.01	1.38	0.82	0.79	0.26	0.51	0.14	0.42
PCYT1A	0.96	1.65	0.65	0.66	0.20	0.10	<0.01	0.10
PEMT	1.60	2.01	1.80	1.66	0.46	0.65	0.79	0.32
Lipid metabolism								
CPT1A	1.04	1.37	1.01	0.66	0.26	0.97	0.16	0.18
DGAT1	1.01	1.21	0.71	0.71	0.22	0.64	0.08	0.66
HMGCS2	1.04	0.52	0.82	0.51	0.25	0.10	0.64	0.69
MTTP	1.00	1.23	1.01	0.88	0.25	0.82	0.48	0.44
PPARα	0.93	0.63	1.16	0.72	0.20	0.08	0.43	0.75

[1]Prepartum, cows were supplemented with 60 g/d RPC (ReaShure, Balchem Corp.; CHO), 12 g/d RPM (Smartamine M, Adisseo USA Inc.; MET), or both (CHO + MET). Postpartum, cows were supplemented with 60 g/d RPC, 18 g/d RPM, or both. Cows were monitored and treatments were applied from 21 d before expected calving through 35 DIM. Prepartum (−21 d relative to calving) gene expression fold-change was used as a covariate in the statistical model.

[2]*BHMT* = betaine-homocysteine S-methyltransferase; *GSS* = glutathione synthetase; *MTR* = 5-methyltetrahydrofolate-homocysteine methyltransferase; *PCYT1A* = phosphate cytidylyltransferase 1, choline, α; *PEMT* = phosphatidylethanolamine N-methyltransferase; *CPT1A* = carnitine palmitoyltransferase 1A; *DGAT1* = diacylglycerol O-acyltransferase 1; *HMGCS2* = 3-hydroxy-3methylglutaryl-CoA synthase 2; *MTTP* = microsomal triglyceride transfer protein; *PPARα* = peroxisome proliferator activated receptor α.

driven by changes in LPC 16:0 and LPC 18:0, where CHO increased their secretion only when fed without MET. Previous studies have shown increases in total plasma LPC (de Veth et al., 2016; Zenobi et al., 2018) as well as individual LPC species in response to feeding protected choline sources (Zenobi et al., 2018; Wang et al., 2021). Although we did not observe an increase in plasma total LPC concentrations in response to CHO, our results did indicate increases in concentrations of LPC 16:0 and 18:1 when CHO was fed without MET for multiparous cows. The fact that MET negated this increase seems indicative of a shift in priority when sufficient Met was available. The tendency for MET to decrease total plasma LPC concentrations in multiparous cows further supports this idea. Assuming that Met supply was sufficient for primiparous cows in our study, this observation would coincide with the lack of effect of CHO or MET on plasma total LPC and LPC 16:0, 18:1, and 18:0 in primiparous cows. The observation for plasma LPC 18:1 concentration aligns with our observation for milk secretion of LPC 18:1 in multiparous cows, in which MET minimized the CHO-induced increase. Research in nonruminants has indicated that LPC has immunomodulatory effects (Liu et al., 2020), which provides a mechanism through which dietary choline or Met could interact with the immune system. Because MET seemed to mute the CHO-induced increases of these LPC species, it is possible that MET may have provided an alternative mechanism for activating the immune system, such as through glutathione (Martinov et al., 2010). In addition to its effects on the immune system, LPC is also associated with energy metabolism. Both LPC 16:0 and LPC 18:1 are shown to be reduced during weight loss in obese humans (Heimerl et al., 2014). The elevation of these LPC species in multiparous cows fed CHO without MET could indicate that these cows were mobilizing less body fat. This idea is supported by the numerically greater body condition score, greater body weight, and lower plasma fatty acids observed for the multiparous cows fed this treatment during the postpartum period (Potts et al., 2020). The increase in LPC secretion in milk and concentration in plasma for cows fed CHO suggests an increase in PC conversion to LPC, which is supportive of a sufficient PC supply. Overall, these findings demonstrate that supplemental choline can affect LPC synthesis, which has the potential to modulate the immune system as well as energy metabolism, but that these changes depend on the Met availability to the animal.

Despite a lack of treatment effect on total PC secretion in milk and concentration in plasma, the changes in secretion of several PC species for multiparous cows suggests that both CHO and MET affected PC synthesis pathways. Research in rodents indicated that SFA- and UFA-PC species are typically derived from the CDP-choline pathway, whereas PUFA-PC species are derived from the PEMT pathway (DeLong et al., 1999). Although these findings can provide valuable insight for interpreting our results, caution should be exercised when extrapolating these findings to ruminant animals, which typically have a relatively limited access to di-

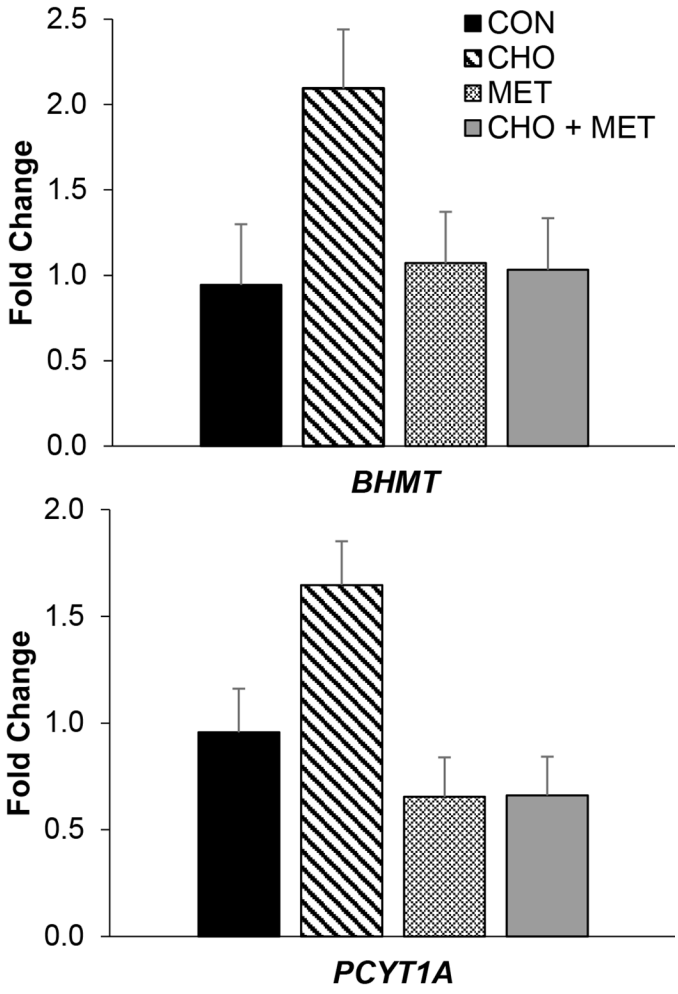


Figure 1. Differences in postpartum hepatic mRNA expression of the genes *BHMT* and *PCYT1A* that are associated with Met and choline metabolism, respectively, in multiparous cows (n = 20) fed the control (CON) diet or the CON diet plus rumen-protected choline (RPC; 60 g/d; CHO), rumen-protected Met (RPM; 12 g/d prepartum, 18 g/d postpartum; MET), or both (CHO + MET) from samples taken ~7 d after calving. Data are expressed as covariate-adjusted fold-changes relative to the CON treatment. For both genes, MET × CHO tended to be significant ($P = 0.09$ and $P = 0.10$, respectively). For *PCYT1A*, the effect of MET was significant ($P < 0.01$).

etary choline in the small intestine. The fact that CHO and MET both enhanced secretion of several PUFA-PC species suggests that both promoted the PEMT pathway in multiparous cows. However, CHO only increased secretion of these PC species during wk 2, suggesting that choline can modulate the PEMT pathway but that this adaptation takes time. By contrast, the apparent increase in the PEMT pathway was not time-dependent for MET, as it increased PUFA-PC secretion, and likely flux through the PEMT pathway, irrespective of week. The PEMT pathway requires SAM to provide 3 methyl groups to produce PC from phosphatidylethanolamine (Cole et al., 2012). It is possible that there was insufficient Met available to support the PEMT pathway via SAM during wk 1 for cows fed CHO. It appears that feeding MET helped to overcome this challenge irrespective of CHO supplementation. Together, these data suggest that, while both CHO and MET have the ability to support PEMT pathway activities, MET may act earlier to enhance availability of PEMT-derived PC-species through the provision of additional SAM. The general lack of treatment effect on the secretion of individual PC species for primiparous cows suggests minimal change to the apparent balance between the PEMT and CDP-choline pathways (DeLong et al., 1999). This finding agrees with the hypothesis that the primiparous cows in our study had a more positive choline balance and is supported by the increase in free choline and betaine secreted into milk when CHO was fed. If the primiparous cows had achieved a positive choline balance, it is likely that PC supply was already sufficient before CHO or MET supplementation and that there was little need to alter the pathways by which PC needs were being met.

There was minimal effect of CHO or MET on the plasma concentrations of individual PC species in multiparous cows, suggesting minimal shift in PC origin (DeLong et al., 1999). However, in primiparous cows, MET reduced plasma concentrations of several species derived from both PEMT and CDP-choline pathways during wk 2, suggesting that MET decreased PC needs during this time. These results are somewhat contradictory to those presented by Zenobi et al. (2018), who showed an increase in plasma concentrations of several PC species of CDP-choline pathway origin when RPC was fed to feed-restricted dry cows. However, previous research in rats suggested that dietary Met alters plasma concentrations of various PC species, specifically by reducing those that contain linoleic acid and PC 16:0/20:4 (Sugiyama et al., 1997), which would suggest a decrease in PC synthesized via the PEMT pathway. Indeed, we also observed a reduction in plasma concentration of several PEMT-PC species in primiparous cows fed MET during wk 2 postpartum, including PC 16:0/20:4. However, the biological significance of this change is unclear.

Research in humans suggests that the PC and LPC profiles differ among the lipoprotein fractions (Wiesner et al., 2009). Thus, the treatment-induced changes in plasma PC and LPC profiles for cows in our study could reflect alterations in the lipoprotein profile. However, we did not measure plasma lipoprotein fractions in our study. Although the measurement of lipoprotein fractions in bovine plasma is more complex than for humans (Grummer et al., 1986), additional research to examine lipoprotein responses to dietary choline and Met supplementation is needed.

Amino Acids

All changes in plasma AA concentrations due to treatment occurred without any changes in protein intake, as DMI was not affected by treatment (Potts et al., 2020).

Of the nonessential AA, only plasma Ser, glutamate, Gln, and Gly were affected by CHO or MET. Contrary to these observations, previous work with periparturient dairy cows by Zhou et al. (2017b) indicated no effect of Met or choline on plasma Ser, glutamate, Gln, or Gly concentrations. Furthermore, they observed that Met increased plasma Ala, aspartate, Asn, Pro, and glutamate, which is also contrary to our results. It is curious that MET decreased plasma Ser for both primi- and multiparous cows in our study, given that Emmanuel and Kennelly (1984) indicated that Met can be used to synthesize Ser to a limited extent in lactating goats and that Ser is required for the first step toward glutathione production from Hcy, as well as for regeneration of 5, 10-methyl-tetrahydrofolate from tetrahydrofolate in the methyltetrahydrofolate pathway (Selhub, 1999). However, because Ser is a nonessential AA and the rate-limiting substrate for glutathione production is Cys (Lu, 2013), it is unlikely that this reduction is indicative of changes in glutathione production. For the same reasons, the decrease in glutamate concentration during wk 1 postpartum for multiparous cows fed CHO without MET are also probably not related to alterations in glutathione production. Instead, these observations are more likely indicative of increased protein synthesis. This aligns with the absence of a treatment effect on hepatic *GSS* expression and the increased milk protein content we observed for multiparous cows fed MET in our study (Potts et al., 2020).

Consistent with previous work (Blum et al., 1999; Zhou et al., 2017b), MET increased plasma Met for both groups of cows, indicating that the Met fed as RPM was absorbed. However, effects of supplemental Met on plasma concentrations of other essential AA in previous studies has been variable. Previous research reported by Zhou et al. (2017b) indicated that Met increased plasma concentrations of several other essential AA in periparturient cows, including Arg, Lys, and Trp. Contrary to these findings, the essential AA affected by MET for multiparous cows in our study were Ile, Leu, Met, and Phe. Previous work (Blum et al., 1999; Preynat et al., 2009) showed that RPM decreased plasma concentrations of the branched-chain AA Leu, Ile, and Val. Interestingly, we observed the opposite, where MET tended to increase concentrations of the branched-chain AA Ile and Leu. Although changes in branched-chain AA concentrations can be indicative of changes in DMI, intake was similar across treatments in our study (Potts et al., 2020). Alternatively, this response could be explained by alterations in muscle AA metabolism because it occurred in absence of a change in DMI. A recent examination of muscle tissue from periparturient dairy cows fed RPM that demonstrated the ability of MET to affect key pathways related to AA and protein metabolism (Thanh et al., 2022) supports this idea. Leucine has been long been shown to be an important mediator of protein metabolism (Stipanuk, 2007). More recently, both Leu and Ile have been shown to stimulate protein synthesis in bovine mammary cells via mammalian target of rapamycin (Appuhamy et al., 2012). Thus, the increase in plasma Ile and Leu concentrations for multiparous cows in our study could help explain the increased milk protein concentration observed for multiparous cows fed MET (Potts et al., 2020).

Liver Gene Expression

The *PCYT1A* gene encodes phosphate cytidylyltransferase 1 choline α, which is the enzyme that catalyzes the rate-limiting step of the CDP-choline pathway (Li and Vance, 2008). Our observations lend support to the hypothesis that supplemental choline increases PC synthesis via the CDP-choline pathway. This change coincided with the increase in milk secretion of several CDP-choline-derived PC species by CHO and is suggestive of the ability for supplemental choline to increase PC synthesis via this pathway in periparturient cows. Because this response seemed to be dependent on MET supplementation, it is possible that inclusion of MET redirected supplemental choline for another purpose. Zhou et al., (2017a) also observed an increase in *PCYT1A* mRNA expression for periparturient cows fed choline, although they indicated that supplemental Met also increased *PCYT1A* mRNA expression. Contrary to our results, previous work by Zhou et al. (2017a) indicated that supplemental choline did not affect expression of *BHMT* for periparturient cows. However, CHO tended to increase *BHMT* expression when fed without MET, suggesting an increase in Met recycling from Hcy via the BHMT pathway for these cows. This response seemed to be dependent on MET supplementation, which could be indicative of a reduced need to synthesize Met from Hcy via this pathway when Met supply is sufficient. In support of this hypothesis, recent in vitro work by Zhang et al. (2016) showed a reduction in *BHMT* mRNA expression with increasing concentrations of Met, indicating that when Met is in ample supply, less Met is regenerated from Hcy through the BHMT pathway.

Lack of an effect of CHO or MET on mRNA expression of *PEMT, GSS*, or *MTR* suggests that the effects

of these nutrients on these pathways do not occur at the transcription level. This observation for *PEMT* expression aligns with the lack of effect of CHO or MET on the plasma concentrations PEMT-derived PC species in multiparous cows. However, this is contrary to previous work which indicated an increase in hepatic mRNA expression of *PEMT* in periparturient cows fed supplemental Met (Preynat et al., 2010; Osorio et al., 2014; Zhou et al., 2017a). It is possible that effects of CHO or MET do not occur at the transcription level for *MTR* or *GSS*. This idea is supported by the work of Zhou et al. (2017a), who reported a decrease in MTR activity, but not *MTR* mRNA expression, for periparturient cows fed supplemental choline or Met. It is also possible that, with only 5 cows per treatment, we did not have sufficient statistical power in our study to detect changes in mRNA expression of these genes.

The mRNA levels of *HMGCS2* and *PPARα* tended to be reduced by CHO (Table 12), which would suggest reductions in ketogenesis and hepatic fatty acid oxidation by cows fed choline. However, CHO did not affect postpartum plasma BHBA or nonesterified fatty acid concentrations for multiparous cows (Potts et al., 2020). Previous research did not show changes in *HMGCS2* (Morrison et al., 2018) or *PPARα* (Goselink et al., 2013; Morrison et al., 2018) mRNA expression when choline was fed to periparturient cows. Feeding MET tended to reduce mRNA expression of *DGAT1* (Table 12), suggesting a reduction in liver triglyceride (**TG**) synthesis, although there was no effect of MET on liver TG content (Potts et al., 2020). The lack of a CHO effect on genes associated with TG synthesis observed in our study and by others (Goselink et al., 2013) suggests that choline has little effect on liver TG synthesis during the periparturient period. Our observations for hepatic gene expression agree with the lack of treatment effect on energy balance, plasma fatty acid concentrations, and liver TG content for multiparous cows in this study (Potts et al., 2020). However, it is worth noting that the multiparous cows in our study did not experience severe negative energy balance during the postpartum period (−3.1 Mcal/d) and that liver TG accumulation was relatively mild (4.9% DM), which may have affected gene expression responses.

Feeding CHO or MET did not affect mRNA levels of *MTTP* in this study. Microsomal triglyceride transfer protein is an enzyme involved in packaging TG into very low density lipoprotein in the liver (Bernabucci et al., 2004). An increase in *MTTP* mRNA expression could indicate an increase in the rate of very low density lipoprotein formation, which is the proposed route by which choline could reduce hepatic TG concentration in dairy cows (Cooke et al., 2007). Previous research has indicated variability in the effect of choline supplementation on *MTTP* expression, where some studies showed an increase (Goselink et al., 2013; Caprarulo et al., 2020) and others (Morrison et al., 2018) showed no effect. Our observations are consistent with those of Morrison et al. (2018) and suggest a limited effect of CHO on TG export for the cows in the current study.

CONCLUSIONS

With the exception of milk secretion of 3 LPC species, effects of choline and Met on apparent choline metabolism were not consistent for primi- and multiparous cows. Although there were subtle changes in the PC profile of milk and plasma in response to supplemental choline or Met, substantial, consistent changes did not suggest definitive shifts in the balance of PC synthesis via the PEMT or CDP-choline pathways. Supplemental choline alone tended to increase hepatic expression of genes associated with choline metabolism in multiparous cows, which suggests a probable role in promoting the CDP-choline and BHMT pathways. However, when supplemental Met was combined with supplemental choline, this response seemed to be negated, which suggests a dependency on Met availability to the cow. This is further supported by the differences in responses between primi- and multiparous cows, who likely differ in their choline and Met requirements. Additional work is required to explore how choline and Met affect one-carbon metabolism during the periparturient period and how these responses may differ with choline and methionine availability to the cow.

ACKNOWLEDGMENTS

Partial funding for this study was provided by Balchem Corporation (New Hampton, NY). The authors acknowledge the staff at the Central Maryland Research and Education Center Dairy Unit (Clarksville, MD) for their assistance with animal care and management. The authors are also grateful to Claudia Gomez and Emily Davis (University of Maryland, College Park) for assisting with sample and data collection. The authors have not stated any conflicts of interest.

REFERENCES

Appuhamy, N. J. A. D. R., N. A. Knoebel, W. A. D. Nayananjalie, J. Escobar, and M. D. Hanigan. 2012. Isoleucine and leucine independently regulate mTOR signaling and protein synthesis in MAC-T cells and bovine mammary tissue slices. J. Nutr. 142:484–491. https://doi.org/10.3945/jn.111.152595.

Artegoitia, V. M., J. L. Middleton, F. M. Harte, S. R. Campagna, and M. J. de Veth. 2014. Choline and choline metabolite patterns and associations in blood and milk during lactation in dairy cows. PLoS One 9:e103412. https://doi.org/10.1371/journal.pone.0103412.

Atkins, K. B., R. A. Erdman, and J. H. Vandersall. 1988. Dietary choline effects on milk yield and duodenal choline flow in dairy cattle. J. Dairy Sci. 71:109–116. https://doi.org/10.3168/jds.S0022-0302(88)79531-4.

Avota, E., M. N. de Lira, and S. Schneider-Schaulies. 2019. Sphingomyelin breakdown in T cells: Role of membrane compartmentalization in T cell signaling and interference by a pathogen. Front. Cell Dev. Biol. 7:152. https://doi.org/10.3389/fcell.2019.00152.

Bernabucci, U., B. Ronchi, L. Basiricò, D. Pirazzi, F. Rueca, N. Lacetera, and A. Nardone. 2004. Abundance of mRNA of apolipoprotein B_{100}, apolipoprotein E, and microsomal triglyceride transfer protein in liver from periparturient dairy cows. J. Dairy Sci. 87:2881–2888. https://doi.org/10.3168/jds.S0022-0302(04)73418-9.

Blum, J. W., R. M. Bruckmaier, and F. Jans. 1999. Rumen-protected methionine fed to dairy cows: Bioavailability and effects on plasma amino acid pattern and plasma metabolite and insulin concentrations. J. Dairy Sci. 82:1991–1998. https://doi.org/10.3168/jds.S0022-0302(99)75435-4.

Caprarulo, V., S. J. Erb, T. L. Chandler, M. G. Zenobi, B. A. Barton, C. R. Staples, and H. M. White. 2020. The effects of prepartum energy intake and peripartum rumen-protected choline supplementation on hepatic genes involved in glucose and lipid metabolism. J. Dairy Sci. 103:11439–11448. https://doi.org/10.3168/jds.2020-18840.

Caudill, M. A. 2010. Pre- and postnatal health: Evidence of increased choline needs. J. Am. Diet. Assoc. 110:1198–1206. https://doi.org/10.1016/j.jada.2010.05.009.

Chang, W., G. M. Hatch, Y. Wang, F. Yu, and M. Wang. 2019. The relationship between phospholipids and insulin resistance: From clinical to experimental studies. J. Cell. Mol. Med. 23:702–710. https://doi.org/10.1111/jcmm.13984.

Chiang, P. K., R. K. Gordon, J. Tal, G. C. Zeng, B. P. Doctor, K. Pardhasaradhi, and P. P. McCann. 1996. S-Adenosylmethionine and methylation. FASEB J. 10:471–480. https://doi.org/10.1096/fasebj.10.4.8647346.

Cole, L. K., J. E. Vance, and D. E. Vance. 2012. Phosphatidylcholine biosynthesis and lipoprotein metabolism. Biochim. Biophys. Acta Mol. Cell Biol. Lipids 1821:754–761. https://doi.org/10.1016/j.bbalip.2011.09.009.

Coleman, D. N., A. Alharthi, V. Lopreiato, E. Trevisi, M. Miura, Y. Pan, and J. J. Loor. 2019a. Choline supply during negative nutrient balance alters hepatic cystathionine β-synthase, intermediates of the methionine cycle and transsulfuration pathway, and liver function in Holstein Cows. J. Dairy Sci. 102:8319–8331. https://doi.org/10.3168/jds.2019-16406.

Coleman, D. N., M. Vailati-Riboni, A. A. Elolimy, F. C. Cardoso, S. L. Rodriguez-Zas, M. Miura, Y. Pan, and J. J. Loor. 2019b. Hepatic betaine-homocysteine methyltransferase and methionine synthase activity and intermediates of the methionine cycle are altered by choline supply during negative energy balance in Holstein cows. J. Dairy Sci. 102:8305–8318. https://doi.org/10.3168/jds.2018-16204.

Cooke, R. F., N. Silva Del Río, D. Z. Caraviello, S. J. Bertics, M. H. Ramos, and R. R. Grummer. 2007. Supplemental choline for prevention and alleviation of fatty liver in dairy cattle. J. Dairy Sci. 90:2413–2418. https://doi.org/10.3168/jds.2006-028.

de Veth, M. J., V. M. Artegoitia, S. R. Campagna, H. Lapierre, F. Harte II, and C. L. Girard. 2016. Choline absorption and evaluation of bioavailability markers when supplementing choline to lactating dairy cows. J. Dairy Sci. 99:9732–9744. https://doi.org/10.3168/jds.2016-11382.

DeLong, C. J., and Y.-J. Shen., MThomas, J., and Z. Cui. 1999. Molecular distinction of phosphatidylcholine synthesis between the CDP-choline pathway and phosphatidyethanolamine methylation pathway. J. Biol. Chem. 274:29683–29688. https://doi.org/10.1074/jbc.274.42.29683.

Duplessis, M., H. Lapierre, D. Pellerin, J.-P. Laforest, and C. L. Girard. 2017. Effects of intramuscular injections of folic acid, vitamin B_{12}, or both, on lactational performance and energy status of multiparous dairy cows. J. Dairy Sci. 100:4051–4064. https://doi.org/10.3168/jds.2016-12381.

Emmanuel, B., and J. J. Kennelly. 1984. Kinetics of methionine and choline and their incorporation into plasma lipids and milk components in lactating goats. J. Dairy Sci. 67:1912–1918. https://doi.org/10.3168/jds.S0022-0302(84)81524-6.

Goselink, R. M. A., J. van Baal, H. C. A. Widjaja, R. A. Dekker, R. L. G. Zom, M. J. de Veth, and A. M. van Vuuren. 2013. Effect of rumen-protected choline supplementation on liver and adipose gene expression during the transition period in dairy cattle. J. Dairy Sci. 96:1102–1116. https://doi.org/10.3168/jds.2012-5396.

Grummer, R. R., W. L. Hurley, C. L. Davis, and C. A. Meacham. 1986. Effect of isolation temperature on the determination of bovine plasma very low density lipoprotein concentrations. J. Dairy Sci. 69:2083–2090. https://doi.org/10.3168/jds.S0022-0302(86)80640-3.

Heimerl, S., M. Fischer, A. Baessler, G. Liebisch, A. Sigruener, S. Wallner, and G. Schmitz. 2014. Alterations of plasma lysophosphatidylcholine species in obesity and weight loss. PLoS One 9:e111348. https://doi.org/10.1371/journal.pone.0111348.

Li, Z., and D. E. Vance. 2008. Thematic Review Series: Glycerolipids. Phosphatidylcholine and choline homeostasis. J. Lipid Res. 49:1187–1194. https://doi.org/10.1194/jlr.R700019-JLR200.

Li, Z., H. Zhang, J. Liu, C.-P. Liang, Y. Li, Y. Li, G. Teitelman, T. Beyer, H. H. Bui, D. A. Peake, Y. Zhang, P. E. Sanders, M.-S. Kuo, T.-S. Park, G. Cao, and X.-C. Jiang. 2011. Reducing plasma membrane sphingomyelin increases insulin sensitivity. Mol. Cell. Biol. 31:4205–4218. https://doi.org/10.1128/MCB.05893-11.

Liu, P., W. Zhu, C. Chen, B. Yan, L. Zhu, X. Chen, and C. Peng. 2020. The mechanisms of lysophosphatidylcholine in the development of diseases. Life Sci. 247:117443. https://doi.org/10.1016/j.lfs.2020.117443.

Lu, S. C. 2013. Glutathione synthesis. Biochim. Biophys. Acta. Gen. Subj. 1830:3143–3153. https://doi.org/10.1016/j.bbagen.2012.09.008.

Martinov, M. V., V. M. Vitvitsky, R. Banerjee, and F. I. Ataullakhanov. 2010. The logic of the hepatic methionine metabolic cycle. Biochim. Biophys. Acta. Proteins Proteomics 1804:89–96. https://doi.org/10.1016/j.bbapap.2009.10.004.

Mentch, S. J., and J. W. Locasale. 2016. One-carbon metabolism and epigenetics: Understanding the specificity. Ann. N. Y. Acad. Sci. 1363:91–98. https://doi.org/10.1111/nyas.12956.

Morrison, E. I., H. Reinhardt, H. Leclerc, T. J. Devries, and S. J. LeBlanc. 2018. Effect of rumen-protected B vitamins and choline supplementation on health, production, and reproduction in transition dairy cows. J. Dairy Sci. 101:9016–9027. https://doi.org/10.3168/jds.2018-14663.

NRC. 2001. Nutrient Requirements of Dairy Cattle, 7th rev. ed. Natl. Acad. Press.

Nixon, G. F. 2009. Sphingolipids in inflammation: Pathological implications and potential therapeutic targets. Br. J. Pharmacol. 158:982–993. https://doi.org/10.1111/j.1476-5381.2009.00281.x.

Osorio, J. S., P. Ji, J. K. Drackley, D. Luchini, and J. J. Loor. 2014. Smartamine M and MetaSmart supplementation during the periparturient period alter hepatic expression of gene networks in 1-carbon metabolism, inflammation, oxidative stress, and the growth hormone-insulin-like growth factor 1 axis pathways. J. Dairy Sci. 97:7451–7464. https://doi.org/10.3168/jds.2014-8680.

Pinotti, L., A. Baldi, and V. Dell'Orto. 2002. Comparative mammalian choline metabolism with emphasis on the high-yielding dairy cow. Nutr. Res. Rev. 15:315–332. https://doi.org/10.1079/NRR200247.

Potts, S. B., C. M. Scholte, K. M. Moyes, and R. A. Erdman. 2020. Production responses to rumen-protected choline and methionine supplemented during the periparturient period differ for primi- and multiparous cows. J. Dairy Sci. 103:6070–6086. https://doi.org/10.3168/jds.2019-17591.

Preynat, A., H. Lapierre, M. C. Thivierge, M. F. Palin, N. Cardinault, J. J. Matte, A. Desrochers, and C. L. Girard. 2010. Effects of supplementary folic acid and vitamin B_{12} on hepatic metabolism of dairy cows according to methionine supply. J. Dairy Sci. 93:2130–2142. https://doi.org/10.3168/jds.2009-2796.

Preynat, A., H. Lapierre, M. C. Thivierge, M. F. Palin, J. J. Matte, A. Desrochers, and C. L. Girard. 2009. Effects of supplements of folic

acid, vitamin B_{12}, and rumen-protected methionine on whole body metabolism of methionine and glucose in lactating dairy cows. J. Dairy Sci. 92:677–689. https://doi.org/10.3168/jds.2008-1525.
Selhub, J. 1999. Homocysteine metabolism. Annu. Rev. Nutr. 19:217–246. https://doi.org/10.1146/annurev.nutr.19.1.217.
Stipanuk, M. H. 2007. Leucine and protein syntehsis: mTOR and beyond. Nutr. Rev. 65:122–129. https://doi.org/10.1111/j.1753-4887.2007.tb00289.x.
Sugiyama, K., A. Yamakawa, A. Kumazawa, and S. Saeki. 1997. Methionine content of dietary proteins affects the molecular species composition of plasma phosphatidylcholine in rats fed a cholesterol-free diet. J. Nutr. 127:600–607. https://doi.org/10.1093/jn/127.4.600.
Sunny, N. E., and B. J. Bequette. 2010. Gluconeogenesis differs in developing chick embryos derived from small compared with typical size broiler breeder eggs. J. Anim. Sci. 88:912–921. https://doi.org/10.2527/jas.2009-2479.
Swartz, T. H., B. J. Bradford, O. Malysheva, M. A. Caudill, L. K. Mamedova, and K. A. Estes. 2022. Effects of dietary rumen-protected choline supplementation on colostrum yields, quality, and choline metabolites from dairy cattle. JDS Commun. 3:296–300. https://doi.org/10.3168/jdsc.2021-0192.
Thanh, L. P., Q. Jiang, N. Wichasit, F. Batistel, C. Parys, J. Guyader, and J. J. Loor. 2022. Alterations in skeletal muscle mRNA abundance in response to ethyl-cellulose rumen-protected methionine during the periparturient period in dairy cows. Animals (Basel) 12:1641. https://doi.org/10.3390/ani12131641.
Wang, F., J. E. Rico, A. B. P. Fontoura, R. Gervais, and J. W. McFadden. 2021. Short communication: Effects of dietary deoiled soy lecithin supplementation on circulating choline and choline metabolites, and the plasma phospholipid profile in Holstein cows fed palm fat. J. Dairy Sci. 104:1838–1845. https://doi.org/10.3168/jds.2020-18798.
Wiesner, P., K. Leidl, A. Boettcher, G. Schmitz, and G. Liebisch. 2009. Lipid profiling of FPLC-separated lipoprotein fractions by electrospray ionization tandem mass spectrometry. J. Lipid Res. 50:574–585. https://doi.org/10.1194/jlr.D800028-JLR200.
Yao, Z., and D. E. Vance. 1988. The active synthesis of phosphatidylcholine is required for very low density lipoprotein secretion from rat hepatocytes. J. Biol. Chem. 263:2998–3004. https://doi.org/10.1016/S0021-9258(18)69166-5.
Zang, Y., S. S. Samii, W. A. Myers, H. R. Bailey, A. N. Davis, E. Grilli, and J. W. McFadden. 2019. Methyl donor supplementation suppresses the progression of liver lipid accumulation while modifying the plasma triacylglycerol lipidome in periparturient Holstein dairy cows. J. Dairy Sci. 102:1224–1236. https://doi.org/10.3168/jds.2018-14727.
Zeisel, S. H., and K.-A. da Costa. 2009. Choline: An essential nutrient for public health. Nutr. Rev. 67:615–623. https://doi.org/10.1111/j.1753-4887.2009.00246.x.
Zenobi, M. G., R. Gardinal, J. E. Zuniga, A. L. G. Dias, C. D. Nelson, J. P. Driver, B. A. Barton, J. E. P. Santos, and C. R. Staples. 2018. Effects of supplementation with ruminally protected choline on performance of multiparous Holstein cows did not depend upon prepartum caloric intake. J. Dairy Sci. 101:1088–1110. https://doi.org/10.3168/jds.2017-13327.
Zhang, Q., S. J. Bertics, N. D. Luchini, and H. M. White. 2016. The effect of increasing concentrations of dl-methionine and 2-hydroxy-4-(methylthio) butanoic acid on hepatic genes controlling methionine regeneration and gluconeogenesis. J. Dairy Sci. 99:8451–8460. https://doi.org/10.3168/jds.2016-11312.
Zhou, Z., O. Bulgari, M. Vailati-Riboni, E. Trevisi, M. A. Ballou, F. C. Cardoso, and D. N. Luchini. 2016a. Rumen-protected methionine compared with rumen-protected choline improves immunometabolic status in dairy cows during the peripartal period. J. Dairy Sci. 99:8956–8969. https://doi.org/10.3168/jds.2016-10986.
Zhou, Z., T. A. Garrow, X. Dong, D. N. Luchini, and J. J. Loor. 2017a. Hepatic activity and transcription of betaine-homocysteine methyltransferase, methionine synthase, and cystathionine synthase in periparturient dairy cows are altered to different extents by supply of methionine and choline. J. Nutr. 147:11–19. https://doi.org/10.3945/jn.116.240234.
Zhou, Z., M. Vailati-Riboni, D. N. Luchini, and J. J. Loor. 2017b. Methionine and choline supply during the periparturient period alter plasma amino acid and one-carbon metabolism profiles to various extents: Potential role in hepatic metabolism and antioxidant status. Nutrients 9:10. https://doi.org/10.3390/nu9010010.
Zhou, Z., M. Vailati-Riboni, E. Trevisi, J. K. Drackley, D. N. Luchini, and J. J. Loor. 2016b. Better postpartal performance in dairy cows supplemented with rumen- protected methionine compared with choline during the peripartal period. J. Dairy Sci. 99:8716–8732. https://doi.org/10.3168/jds.2015-10525.

ORCIDS

S. B. Potts https://orcid.org/0000-0002-4952-4193
K. M. Moyes https://orcid.org/0000-0003-3591-3849
R. A. Erdman https://orcid.org/0000-0001-6954-4282